

Homework 3: (Return by Sunday 20.10.25 in PDF -format)

Design a lossless RLCG-simulation model for a $3 \cdot X$ inches long 60Ω transmission line. The rise time of the signal is $1.0 \text{ ns} \cdot X$, the circuit board's $\delta r = 4.5$ (1 mils = 1/1000 inch and 1 inch = 25.4 mm). $X = 1 + n/5$, where n is the last digit of your student number. A cross-section of the transmission line is shown in Fig. 1. Show with simulation results that the model you designed works correctly: compare the model you built from inductors and capacitors to an ideal transmission line found in LTspice ("tline" in LTspice). Experiment with the model and tell what effect the number of RLCG-elements have? Help for the solution of the homework can be found in the lecture material.

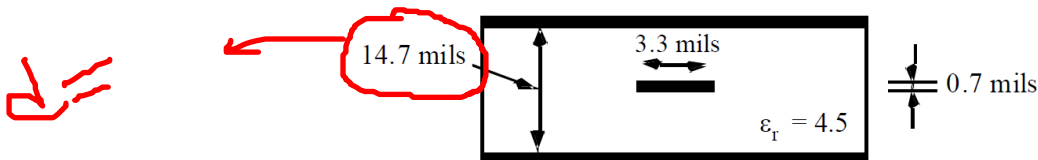


Fig. 1. A cross-section of the transmission line.

The simulations can be done with the simulator found in <https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>. You can find quick guides to LTspice online. Questions to jaakko.huikari@oulu.fi.

Important practical notes on your return documents on this course:

- Name your return file in the following manner: ESD24 HW#N first name surname. As an example, in the case of homework 1 => ESD24 HW#1 Jaakko Huikari (first- /surname matching your name in the course's Moodle page (Enrolled users)).
- Return your documents in **PDF -format**. Do not return documents in Word- or any other formats. Convert your word-document, for example, into PDF -format before you submit it.
- Place your name and student ID number at the top edge of your return document. Use the name matching your name in the course's Moodle page (Enrolled users).
- Make sure your answers and simulation results are clearly visible in your return documents. If necessary, use cursors, zoom and screen captures to present sufficient graphic information (waveforms etc.). Make sure that figure axes are also visible in your return documents.
- Present the reasoning behind your answers/ results including theoretical basis and equations etc. (found in the lecture material). Presenting mere outcome is inadequate answer.

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